

INFORMATION SYSTEMS SECURITY

Definition: Information systems security is the policies, procedures and technical measures used to prevent unauthorised access, alteration, theft or physical damage to information systems. It involves the safeguards required to protect info systems against threats/hazards which might cause unauthorized modification, disclosure, destruction of data/information or software. Security controls should be put in place to protect computer based-information systems (hardware, software, live ware and database) against the hazards to which computerized information systems are exposed. Security controls help assure high systems standards and performance by protecting the system against hardware, software, and live ware failure.

Protection should cover the following areas:-

1. Data and information from hazards and abuse.
2. Unauthorized use of the computer and allied resources.
3. Accidental alteration of data, programs, and information.
4. Privacy of data, information and programs.
5. To ensure both hardware and software have longer life span.

Environmental threats/hazards to information systems resources.

1. Fire – Is the most serious and costly hazard as it utterly destroys everything i.e. data, information, software, hardware, live ware, ancillary equipment and the premises. To fight/avoid fire:-
 - a. Use halon gas fire extinguishers, as carbon dioxide extinguishers will endanger any trapped personnel and the water extinguishers will damage the hardware and storage media.
 - b. Store storage media in fire proof cabinets and lockable metal boxes.
 - c. Install smoke detectors.
 - d. Training staff on fire fighting skills.
 - e. Strict observation of safety procedures e.g. avoiding smoking in computer installations premises.
 - f. Place exit signs in conspicuous positions.
2. Water/Flood and moisture. Causes the metallic components to rust. Floods cause serious damage to hardware, network cables and wiring. Security measures against this hazards includes:-
 - a. Setting up computer rooms on higher grounds to avoid floods. Avoid installation of the information system in the basement especially in flood prone areas.
 - b. Adequate drainage system.
 - c. Use of water proof ceilings and floors.
3. Other natural disasters. Includes cyclones, hurricanes, lightening and earthquakes. Protection against natural disasters should be considered when choosing the location of the computer information system.
 - a. Avoid areas prone to floods, lightening and tremors.
 - b. Erect lightening arrestors.

4. Excessive heat or temperature. Excessive heat or temperature from the computer itself or outside the environment can destroy computer storage media or devices. Security measures:-
 - a. Efficient ventilation system.
 - b. Installing cooling systems in the computer rooms e.g. use of fans and air conditioners.
5. Civil strife and Terrorism. Terrorist attack includes activities like political terrorists e.g. bombs, criminal types of activities, hooliganism, individuals with grudges and people intending to cause general disruption. Security measure includes:-
 - a. Control physical access to the premises housing the info system i.e. comp labs, comp rooms, computer resource centre, IT centre, the name various from one organisation to another.
 - b. Avoided terrorism and hooliganism triggering activities e.g. exploitation of workers.
 - c. Consult with police and fire authorities/brigade about potential risks and co-operate with them conscientiously.
6. Sabotage. It is one of the greatest physical risks to computer installations. Saboteurs can do enormous damage to computerized info systems with little risk of apprehension e.g. bomb can be planted, a communication line can be cut or disconnected, deadly computer virus can be deliberately introduced etc. Providing adequate security against such acts of sabotage is extremely difficult and expensive.
7. People threats. These include carelessness, clumsiness and accidental destruction/erasure of data, info or programs as well as theft and piracy of data & software. Security measures:-
 - a. Have a good office layout e.g. such that the supervisor can see what the people under him are doing.
 - b. Limit access to sensitive organisation information.
 - c. Have lockable and disk locks.
 - d. Terminate/dismiss employee with a penchant of misusing the computer resources or re-locate them to another department.
 - e. Careful recruitment of staff.
8. Environmental problems/conditions. More often than not computers are housed in premises that were not originally meant to contain them. This can bring about environmental problems e.g. water and steam pipes may run near or thro' a computer lab, where bursting pipes could cause extensive/considerable damage to the info system. Database on storage media can be destroyed by magnetic fields emanating from electric motors in the vicinity. External radiation, smoke and dust from the surroundings are also hazardous to comp info systems especially storage media. Environmental problems include brownouts (temporary power surges (drops)), power spikes, power failures, Measures to counteract these:-
 - a. Locate IT centre away from this threats.
 - b. Thorough and regular cleaning of the computer labs.

- c. Covering the computer and allied resources with water proof plastic dust covers when they are not in use.
 - d. Have dust mats, double door and monitoring devices to prevent entry of dust,
 - e. General building safeguards, use slab for the walls, water proof ceiling and roofs and easy to clean floor.
 - f. Use UPS, power stabilizers, surge protectors, voltage regulators etc to protect the h/ware from power fluctuations.
9. Computer virus. This is a computer program developed by malicious computer programmers for destructive or disruptive purposes. They are hidden instructions that become attached to other programs or files and cause them to malfunction. It's a rogue program that spreads rampantly through computer systems, destroying data or causing the system to become congested and malfunction. Examples of computer viruses; Melissa, Nimba, Code Red, Sircam, Love Bug, Egerton Lab, Amoeba, Friday the 13th, etc. They can be passed from one computer to another (spread) thro' secondary storage media, thro' data transmission in computer networks, downloading affected files from the internet, and thro' e-mails or malicious employees can knowingly or unknowingly introduce virus in an organizations information system. Other sources of virus spread include; pirated software, games programs, freeware & shareware. Security measures to curb virus threat:-
- a. Install anti-virus software in the info system and scan storage media regularly. Examples of anti-virus MacAfee, Kaspersky, Dr Solomon's anti-virus toolkit, Norton anti-virus,
 - b. Avoid freeware and shareware which have been prime entry point for viruses.
 - c. Storage media for the organisation should not be used elsewhere and private media should not be used in the organisations computer systems, unless under special circumstances and express authority is granted by the ICT depart top management.
 - d. Make backup copies as soon as you open new software package and store the copies off-site.
 - e. Quarantine each new piece of software on an isolated computer and review it carefully before installing it on a network.
 - f. Restrict access to info system resources to only the authorised personnel.
 - g. Check all programs regularly for change of size, which could be a sign of tampering or virus infiltration.
 - h. Institute a plan for immediate removal of all copies of suspicious programs and back up of related data.
 - i. Make sure all purchased software is in its original wrapping or sealed-disk container.
 - j. Install firewall. Firewall is a device that stands in btw the organisations network and the internet checking any type of incoming and outgoing traffic that could potentially damage the network & the organisation's information system. It stops the viruses in the internet finding way into the organisation's computer information system.

10. Cyber crime. Computer criminality is another threat to computer info system security especially software. Computer crime takes different forms. This includes outright theft of cables, h/ware, storage media etc., software piracy, theft of computer time, financial fraud, funds embezzlement, unauthorized alteration of programs and data, theft of computer time, data/info theft (commercial espionage), and malicious destruction of database/hardware or software, unauthorised access (hacking and cracking).

Note: **Hackers** gain unauthorised access to computer systems mainly for the fun and thrill of it whilst **Crackers** gain unauthorised access to computer systems in order to perpetrate theft for profit, for criminal mischief or for destruction.

Physical security:

This includes:-

1. Hiring of security guards to control access to the building housing the info system.
2. Using keyboard locks on terminals used by authorised users.
3. Use of disk locks and disk banks.
4. Strengthening computer labs by putting strong window grills and metallic doors.
5. Installation of electronic surveillance system. This is used by the supervisors of the computer and other electronic equipment to monitor worker's performance often without their knowledge.
6. Data encryption. Data is often sent in coded or encrypted, form over communications lines to keep it secure from unauthorised eyes. Encryption is the encoding of data by converting the standard computer code into a secret code for transmission. After delivery to the destination the data is converted back (decrypted) into standard computer code.
7. Use of passwords – a password is a secret code (word, numbers, symbols or a combination) that must be typed in to gain access to a computer system, a program or a database/file. An effective well-designed password is that which can not easily be guessed by hackers or crackers trying to break into a computerized information system.
8. Biometrics – Is the use of individual body characteristics instead of passwords to gain entry into the computer system e.g. finger print, kiss on screen, lip prints signature, photograph etc.
9. Set up a clear and firm management system security policy on crimes and frauds.
10. Monitor and investigate error logs and reports on regular basis.
11. Carry out risk analysis to examine the organisation's exposure to possible fraud.
12. If possible let there be only one entrance and one exit for ICT centre with trained security guard(s) stationed there.
13. Establish effective security procedures e.g. for regular back up software and database.
14. Paper shredders. Sensitive reports should never be disposed off by simply being thrown in waste containers.
15. Obfuscation – means to isolate by confusing, bewildering, obscuring or hiding something from a potential penetrate.

COMPUTER SYSTEM CONTROLS

Any information system or data processing system, whether computerized or not, must incorporate controls to ensure that it works efficiently, i.e. it must:

- a) Process all the required data accurately.
- b) Provide for continuity and change.
- c) Contain sufficient controls to satisfy the auditors.
- d) Provide the required amount of data security.

In a computer environment four elements must be controlled i.e. Source document, Input, Processing and Output.

Computer system controls can be divided into 3 main areas:

1. Administrative controls.
2. System development controls.
3. Procedural and processing controls.

1. Administrative Controls.

Administrative controls are the responsibility of the management whereas procedural controls are the concern of the system analyst, and system development controls should be decided jointly btw system analyst and the management.

These controls ensure that proper procedures are followed to maintain discipline and efficiency over the day-to-day running of the computer department.

An efficient administrative control system will involve the following aspects:

- (i) Separation of duties.
- (ii) Control over personnel running the computer department.
- (iii) File control.
- (iv) Security against fire hazards.
- (v) Stand-by arrangements and insurance.

Administrative controls can be divided into 2:

(a) **Organizational Controls**

These are necessary because a new function may be introduced into the structure of the organization, which concentrates data processing in one area, whereas previously many user departments could have been involved.

To minimize this problem, a Distributed Data Processing system is used, whereby minicomputers or microcomputers carrying out routine data processing are set up at remote locations, but linked to a central computer.

(b) **Operations Controls**

These are methods and procedures established to:

- (i) Ensure that operations staff follow the correct procedures.
- (ii) Provide adequate arrangements for the physical security of records and files.

The most important points of the Operations control are that:

- 1) Access to source documents should be restricted to data preparation and control staff.
- 2) Access to the computer should be restricted to operation staff.
- 3) Computer operators and programmers must not originate live entries.
- 4) Control staff should not have other EDP duties.

In order to make the controls more effective, the functions of data processing staff, operators, control clerks and librarians must be properly specified and documented.

The proper development and maintenance of operations control will help to: -

- (i) Prevent or detect accidental errors which may occur during processing.
- (ii) Prevent or detect fraudulent manipulation of data during processing.
- (iii) Prevent access to and misuse of personal or otherwise classified information.
- (iv) Prevent the destruction of records.

A computer control department/section must be established. This department must be independent of other operating functions. The main function of this department is to ensure that data is processed accurately and completely.

2. System Development Controls

Systems development controls should be decided jointly between the System analyst and the management.

It is important to lay down high standards of control. This is because of the time spent on systems development, the cost involved, and the probable complexity and volume of detail involved.

The main objectives of systems development control are:

- (i) To ensure that an application is computerized only if it is beneficial to do so.
- (ii) To establish a basis for management review and understanding of the system.
- (iii) To ensure the development of effective systems and programs, and reduce wasted programming effort.
- (iv) To ensure that systems and programs are effectively maintained.
- (v) To ensure that proper and complete documentation of the system is created and maintained.

These controls cover such areas as the development, implementation and maintenance of all System and Application software.

The main requirements and techniques of systems development control are:

- (1) Standard procedures and documentation.
- (2) File conversion.
- (3) Review procedures.
- (4) System and program amendments.

3. Procedural and Processing Controls

Procedural controls are normally the concern of the System analyst.

Procedural controls are all those controls exercised over a particular application:

- (a) Input controls.
- (b) Processing and File controls.

- (c) Hardware controls.
- (d) Software (program) controls.
- (e) Output controls.

Sometimes, the term “Processing controls” is used to describe all the controls applied in the Data processing cycle.

Note that, some controls will be performed clerically, while others are performed by the computer itself, and are known as Program or Software controls.

The aim of procedural control is to:

- (i) Ensure that the data processing is complete, e.g. by the setting up of batch controls and subsequent reconciliations by the computer and manually.
- (ii) Ensure the accuracy of data processing, e.g., by proper setting up of well-designed source documents, verification, program validation, etc.
- (iii) Ensure that only authorized data is processed, e.g., by manually authorizing input, restricted access to computer files.
- (iv) Ensure that proper management and audit trails are laid down.

The system should have maximum safeguards against error and fraud in order to prevent:

- (1) Incorrect recording of data at the point of document creation.
- (2) Loss of data in handling.
- (3) Introduction of unauthorized data.
- (4) Incorrect conversion into machine-sensible form.
- (5) Incorrect processing.
- (6) Incorrect utilization of computer Input.

(a) Input Controls

Input controls should be established efficiently.

The main objectives of Input controls are to ensure that:

- (i) All source documents are correctly completed and are transmitted to the computer department.
- (ii) All source documents are received by the computer department and are correctly converted into the computer input media (e.g. Magnetic tape, floppy disc, etc).
- (iii) All the data on the input medium is transmitted accurately to the computer centre and accepted by the computer.

(b) Processing and File Controls

Controls must be applied to ensure the accuracy of processing the data. These will include:

- (i) Hardware controls.
- (ii) File controls.
- (iii) Software controls.

(c) Hardware Controls

All equipment must be properly maintained and kept in the recommended environment.

Magnetic tape and disc systems require fairly rigid temperature, humidity and power specifications, and therefore control devices should be used to detect variations from the norm.

The equipment itself should have proper hardware controls, which may include:

- (i) Parity checks.
- (ii) Overflow checks.
- (iii) Validity checks.
- (iv) Printer timing checks.
- (v) Terminal readiness checks.
- (vi) Data transmission checks.
- (vii) Keeping of daily computer logs.

4. File Controls

To ensure that only the right files are used in processing,

- (i) Proper file conversion controls must be exercised during Systems implementation.
- (ii) All staff should be properly trained and provided with operation manuals.
- (iii) Computer files should be controlled by a Librarian.
- (iv) Computer files should have external file labels.
- (v) Computer files should have internal file labels.

Note. The Input file labels can be checked by the computer operating systems. They comprise of:

- (1). **Header records**, which contain;
 - (i) Label identifiers.
 - (ii) File identifier.
 - (iii) Generation/run number of creation.
 - (iv) Date of creation.
 - (v) Retention period.
- (2). **Trailer records**, which contain;
 - (i) End-of-file marker.
 - (ii) Control totals.

(d) Software (Program) Controls

To ensure that only valid data is processed, all input should undergo validation tests.

The Data vet program may be applied to batches of data or to individual transactions (in an Online or Real-time system), and any erroneous records rejected and reported (on a VDU in online input).

Software control procedures should detect some errors immediately after they occur.

The following is a brief classification of the most commonly used checks;

- (i) Automatic file label checking.
- (ii) Read-after-write check to compare data written in an output record with that retained in the Main memory.
- (iii) Address comparison check to compare address location where data is held to that specified by the program instruction.
- (iv) Arithmetic checks on the accuracy of a calculation.

- (v) Overflow checks to ensure that, if the result of a calculation cannot be fitted in the address allocated, the overflow part will not be lost.
- (vi) Input/Output Control (IOCS) check. IOCS is a software package supplied by the manufacturers. They can also be developed by individual users. This checks the accuracy of input and output.

(e) Output Controls

The computer should be programmed to write Page numbers, descriptive headings, and processing dates on all print outs.

Output controls are designed to ensure that:

- (i) All output is received from the computer.
- (ii) Where possible, the Output is reconciled with input. Control totals can be manually reconciled by the control/user departments or by the computer.
- (iii) All output and the rejection and exception reports are acted on.
Proper procedures should exist for error correction and re-input, e.g. every error should be logged, and cleared on re-input.
- (iv) Output is correctly distributed and its distribution logged.

Master File Controls

There are 3 important aspects in Master file control;

- (1). Amendment to standing data.
- (2). Maintenance of standing data.
- (3). Maintenance of transaction data.

The degree of control required for standing data is more than that for transaction data.

Once standing data is entered into master files, it is important that there are adequate controls to ensure that the data remain unaltered until an authorized change is made.

This can be achieved by;

- Periodically printing out of standing data so as to check with clerically held information.
- Establishing independent control totals for periodic verification with equivalent totals accumulated in the files.
- Using the computer programs to establish and verify the control totals.

The degree of control required to maintain transaction data on master files varies according to the contents of the file.

5. Library Controls

In order to adequately operate a tape library, procedures and controls must be established and maintained if maximum and efficient usage is to be realized from Magnetic tapes.

For efficient tape library controls, the following should be provided:

- A quick means for physically locating any reel of tape in the library.
- Since the tape file may have more than one reel, each reel should be identified with the file number and reel number.

- If a file or reel is missing from the library, the name of the person to whom it was issued should be recorded.
- Identify the expiry of the scratch date so that tape reels may be released for reuse. A **Scratch date** is the date that the data recorded on a particular tape is no longer required.
- Maintenance of records should be written in short, easily understood terms, and requires a minimum number of entries.

Note: Library controls for Magnetic tapes may be setup in various ways as long as they provide the proper control of all tape reels in the computer center. However, each time the tape status changes, all records belonging to that tape must be changed accordingly.

Other types of controls

1. Organisational controls
2. Documentation controls
3. Management controls
4. Sociological controls
5. Technological controls
6. Environmental controls

For further reading on computer controls:-

1. System Analysis and Design by Donald Yeates and Tony Wakefield page 279 - 290
2. A text book of Systems Analysis & Design by Alphonse Riaga page 207 – 219
3. Systems Theory and Management Information Systems simplified by Nisar A. Saleemi page 361 – 366

Exercise/Assignment

1. (a) What are computer controls?
(b) State the main purposes and types of computer controls.
2. Describe the salient features of Administrative computer controls.
3. Write short notes on the following:
 - (a) System development control.
 - (b) Procedural controls.
4. Briefly explain the following:
 - (a) Hardware controls.
 - (b) File controls.
 - (c) Software controls.
 - (d) Library controls.

AUDITING COMPUTERIZED INFORMATION SYSTEMS

Auditing: Defn: - Involves having an expert who is not involved in setting up or using a system, examine information in order to ascertain its reliability. Auditing is also a security measure.

Objectives/Aims of auditing a computer-based information system

1. To ensure the quality of the information contained in/produced by the system i.e. the aim is quality assurance of information.
2. To ensure that the system is working properly and that the (financial) information can be relied upon.
3. To review and evaluate whether proper and adequate data processing controls, administrative controls and procedural controls have been developed and implemented.
4. To determine whether resources are being used in a cost/effective manner.
5. To check that assets are properly safeguarded and not used improperly.
6. To review integrity, reliability and efficiency of the information system and the financial reports it produces.
7. To establish the image of the computer as a corporate aid for use by any department that needs it.
8. To maintain proper managerial controls of computer resources and EDP activities.
9. To include a test of accuracy and integrity of the data processing of several important computer applications.
10. To ensure that the reporting structure of the ICT department is proper.
11. To reduce or eliminate the possibility of fraud or errors in the information system.

2 types of Audit/Auditors

1. Internal Audits

Aim to prevent fraud by evaluating internal (financial) control systems, and are a useful resource to help in the development of control procedures for information systems.

- Done by internal Auditors, who are employees of the organization that owns the I.S.
- Internal auditors study the controls used in the information system to make sure that they are adequate & that they are doing what they are purported to be doing/supposed to do.
- Internal auditors also test the adequacy of security controls.
- Though they work for the same organization, internal auditors do not report to the people responsible for the system they are auditing.
- Their work is often more in-depth than that of external auditors.

2. External Audit

- Gives an independent, reliable and expert opinion about financial statements: Do the financial statements, give a true and fair view/reflection of the state of the business/organization?
- Done by external auditors, who audit the system to ensure the fairness of the financial statements being produces.

- They may also be brought in if there is something out of the ordinary accruing that involves organization employees, e.g. suspected computer fraud or funds embezzlement.
- Auditors may require that certain self-checking mechanisms and authorization procedures be incorporated in organization/systems, especially those supporting pension funds, or financial institutions e.g. banks are obvious examples.

NB: It is very much better & easier, if these audit requirements can be taken into account at the specification & design stage, rather than after the system is complete, so the analyst must talk to all the relevant authorities & find out their requirements alongside those of the more obvious users of the system.

3 methods/ways of auditing Computer-Based Information Systems (CB-MIS)

Three common methods of auditing in a computer.

1. Auditing around the computer.
2. Auditing through the computer
3. Auditing with the computer

1. Auditing around the computer

- No attempt is made to check the internal processes of the system. The auditor concentrates on the inputs & outputs and checks that the outputs derived from the inputs are correct according to the procedures carried out. This technique/method/way/type of audit regards the computer processing as taking place in a 'black box'.
- Suitable for relatively simple systems. The computer is treated as a 'black box'. Auditor looks at input & output but ignores what goes on in between.
- Began in early days when the auditor was unfamiliar with the comp terminologies, programming & the controls used in a computer system. In this method the comp is viewed as a 'black' box & the auditor do not review/examine the comp system & programs but review/examine the input and output documents only.
- The controls & procedures are considered unimportant as long as the output generated could be traced back to the input and the input was deemed valid.
- The auditors select input & test them against the appropriate output and vice versa. If they match and proved to be accurate it is assumed that the system of controls is operational and that it is working properly.

Advantages of auditing around the computer.

1. There is little risk in tampering with live data.
2. Little technical training of the auditor is required. The auditor need not be computer skilled.
3. It is simple, straight forward & easily understood.
4. The costs of audit resources are generally low.

Disadvantages

1. Many computerised systems are voluminous for proper manual testing.
2. No means are provided by which the auditors get involved and gain firm understanding of the computer system.
3. It ignores the system of controls & thereby fails to recognize potential errors or weaknesses within the system.
4. It represents after the fact rather than preventive auditing.
5. It makes no use of the most powerful and valuable audit tool, the computer.
6. To all intents and purposes it does not achieve the auditor's goals [adequately].

2. Auditing through the computer.

- Checks the external inputs & outputs & the internal processes. Computer audit s/ware packages are usually used to do this.
- With increasing complexity of Info Systems & the greater extent to which they are used in business organizations this technique/method/way/type of audit is now common place.

This technique places a greater emphasis on testing the comp system that produces the output rather than testing the output itself.

The auditor tests & verifies:-

- a) The effectiveness of control procedures over comp operations & computer programs.
- b) The correctness of internal processing.

This audit technique requires that two basics are accomplished. They are:-

- a) The review of and verification of source transactions.
- b) The actual testing of computer programs, logic and programmed controls.

This technique should be thought of as a testing of the program's logic rather than a test of the computer's accuracy.

One of the key tools in applying this technique is the preparation by the auditor of a series of test transactions normally referred to as TESTDECK; this test deck should include both valid and invalid transactions.

- The test deck is run on the computer using the same programs that were used to operate a particular application that is being tested.

Advantages of auditing through the computer

1. It helps the auditor become more involved in the system, thereby increasing his knowledge & ability to perform more complex audits in the future.
2. It works as an aid in making compliance tests and an aid in the evaluation of programmed controls.
3. It increases the service to clients – because controls and operations are checked or at least observed/examined by the auditor.

4. The test results are readily identifiable and can be used as a measure of internal processing reliability.
5. It utilizes the computer as a tool for performing audit functions.

Disadvantages of auditing through the computer.

1. It requires computer time.
2. It requires more technical knowledge and therefore more skilled personnel.
3. It represents after the fact testing rather than preventive testing.
4. It represents only a limited test of the system test deck.

3. Auditing with the computer

- All audits including computerized accounting records should be executed in accordance with the Auditing Standards & with the guidance of the auditing guidelines.
- Computer Assisted Audit Techniques (CAAT) have been developed to overcome the problems of auditing computerized systems.

Auditors can use the computer to assist them in various auditing tasks. Where an accounting information system has been computerized, infact, auditing with the computer is virtually mandatory because the accounting data are stored on the computer storage media and manual access is impossible.

There are audit packages which are used to do this type of auditing or an in-house program can be developed for the purpose.

Advantages of auditing with the computer.

1. It uses computer as an auditing tool.
2. It is very fast
3. It keeps the auditors abreast with the computer technology.
4. It is a more thorough auditing system
5. It represents preventive testing.

Disadvantages

1. Demands technical knowledge of computers
2. Requires computer time.

2 Main/Principal CAATs

1. Use of 'test data' to test the operation of the client's programs.
2. Use of audit s/ware - comp programs developed for audit purposes to examine the contents & do work on the contents of the client's computer files.

Problems/challenges of auditing computerised systems.

1. Are complex & difficult to understand, partly because of the specialist vocabulary used. Larger audit firms have set up specialist sections of computer auditors. The relationship between general audit staff & computer audit staff has to be defined.
2. EDP/Computerized systems are much more complex

3. Technical language is used
4. Many controls are program controls
5. Frequent changes are made to information systems and programs.
6. Too much documentation is available.
7. Lack of visible evidence. Data is stored on magnetic and optical media. The information contained in the records is not easily examined. This is an obvious problem for the auditor.
8. Many computerized systems produce acres of print-outs & the auditor may well be faced with a surfeit of record rather than a shortage.
9. An Auditor can fully understand a manual system in a matter of hours at most. A computerized system cannot usually be fully comprehended without expert knowledge & a great deal of time. (Lack of computer literacy on part of Auditors).
10. If storage of comp. files holding customer's statements of accounts is not maintained properly, it may be difficult & time consuming for the Auditor to access them.
11. Lack of knowledge of computer terms and concepts.
12. I.T. personnel responsible for developing the info systems may have little or no knowledge about accountancy, auditing or the legal framework within which their systems are to operate; this may lead to problems.
13. It is difficult to follow an audit trail in computerized info systems becoz the data within a computer system is 'invisible' to the human being as it is kept in machine-sensible form.

Solutions/remedies to these problems

1. The auditor should acquire working knowledge of computerized info systems & its ramifications. At least one member of the audit team should have extensive first-hand experience of computing.
2. A sound working relationship btw the I.T. staff and the auditors of computerized systems.
3. Auditors should be actively involved in system devt and system modifications especially in the following stages
 - a. Project feasibility study
 - b. Systems design
 - c. Program testing and system testing
 - d. Implementation of the system
4. Auditors should insist on a strict segregation of duties, as in I.T. organizational chart to prevent collusion.
5. Employ 'auditing round the computer' approach. Thus treating the computer as a 'black box'. The auditor is not concerned with what happens inside the computer. This approach requires less computer knowledge on the part of the auditor and may be effective for simple systems.
6. An audit trail may be created where one does not exist. Print outs can be requested on a regular or an ad-hoc basis, either of exceptional items or at random. This will allow the auditors to check that transactions are correctly acted on.
7. Standard audit packages are available which:-

- a. allow random sampling of contents of a file, perhaps printing every nth record.
 - b. cause exceptional items to be printed.
 - c. give control totals of certain important fields. When such packages are not available, they can be developed internally/in-house.
8. Spot checks on the computer installation, and on certain systems which run on the computer, should be made. These will allow the auditors to ensure that standards are being adhered to in all areas – analysis, programming, operations, the library, data preparation & data control – and that, in particular, documentation standards are being enforced.
9. Questionnaires and checklists can be used by the auditors to ensure that all aspects of control are covered.
10. Throughout their work, the auditors must ask questions. Experience will often allow them to sense whether or not something is wrong with a system.

Audit trail. Definition

1. It is the path which a transaction traces through a data processing system from source documents to summary reports. It refers to the facilities or procedures which allow a transaction to be traced through all stages of data processing beginning with its transformation into info on a final output document.
2. Involves tracing/trailing/following the outputs from the inputs especially in an accounting information system to ascertain proper recording, posting and accepted accounting procedures were followed in production of final management accounts.
3. An Audit trail can also refer to a careful study of an information system by experts (auditors, analyst/project managers/system designers/developers) in order to establish or find out all the weaknesses in the system that could lead to security threats and weak access points for cyber criminals. An audit of the information system may seek to answer the following questions.
 - a) Is the information system meeting all its original intended design objectives?
 - b) Have all the security measures been put in place to reduce the risk of computer crimes?
 - c) Are the computers secured in physically restricted areas?
 - d) Is there backup for data and information of the system that can ensure continuity of services even when something serious happens to the current system?
 - e) What real risks face the system at present or in future?

Audit trail consist of such things as reference numbers, dates and names which are recorded in files, ledgers and journals to facilitate the tracking of these records to source documents or to record in other files.

Audit trails are tools designed to help the management in four ways

1. To provide access to info in order to answer inquiries of customers and others.
2. To provide a means to trace and verify transaction detail involving asset control such as disbursements.

3. To provide a means to analyse and compare summarised operating results compiled within the structure of the system.
4. To provide proof of the facts required to establish validity of accounting transactions to meet tax and other legal requirement through trails to communication media and other records or studies.

The auditor uses audit trails which the management has found necessary for internal purposes.

There is lack of audit trail when:-

- a) The comp generates totals, analysis and balances without printing out details.
- b) Exception reports are produced e.g. on overdue accounts without any assurance, apart from a knowledge of the correctness of the program, that the list is complete.

Techniques used to overcome loss of audit trail

- a) Special print outs for auditors. The auditor may request the management to print out the statements required to reconstruct the audit trail with the help of the computer.
- b) Maintenance of file on 'grandfather, father, and son principle' can help to show the auditor the building up of the computer produced balances.
- c) Programmed interrogation facilities
- d) Clerical re-creation
- e) Total testing & comparison with other data, budgets, previous periods.
- f) Alternative tests e.g. physical stock counts.
- g) The use of test packs to verify program performance.

Planning the audit in a computer environment.

All audits must be planned.

1. Auditors need to be involved in computerised systems at the planning, development & implementation stages. Knowledge of the system gained at these stages will enable the auditor to plan his audit with an understanding of the system.
2. Timing is more important in computerised systems than in manual ones. The auditor need to be present when data & files are available. More frequent visits to the client are usually required.
3. Recording methods involves use of secondary storage media.
4. The allocation of suitably qualified staff to audit.
5. The extent to which CAATS can be used also require planning in advance.
Some Audit firms have setup special comp audit departments to audit computerised systems.

Duties of computer audit staff.

1. To develop a working relationship with the client who is installing an EDP/computerized info system.
2. To undertake the system evaluation & write the audit program.
3. To device & carryout the special auditing techniques.
4. To supervise the general audit staff in areas where the special expertise of the comp audit development is needed.

Computer audit programs

- Consists of computer programs used by an Auditor to:-
 1. Read disks & tape files & extract specified information from the files.
 2. Carry out audit work on the contents of the file.
- These programs are sometimes known as enquiry or interrogation programs. Are usually coded in HLLs e.g. COBOL, PASCAL. Are coded by or for an audit firm but client's own interrogation programs can be used; such programs are available from s/ware houses.
- Staff unskilled in programming can be easily taught to put their search or operating requirements into a simple form which the comp audit program can interpret & apply to the files selected.

Uses of computer audit programs

1. Selection of representative or randomly chosen transactions or items for audit tests, e.g. Item number 36 and every 14th item thereafter.
2. Scrutiny of files & selection of exceptional items for examination e.g. all wages payments over £120, or all stock lines worth more than £1,000 in total.
3. Comparison of two files and printing out differences e.g. payrolls at two selected dates.
4. Preparation of exception reports e.g. overdue debts.
5. Stratification of data e.g. stock lines or debtors, with a view to examination only to material items.
6. Carrying out detail tests and calculations
7. Verifying data such as stock or fixed assets at the interim stage & then comparing the examined file with the year end file so that only changed items need be examined at the final audit.
8. Comparison of files at succeeding year ends e.g. to identify changes in the composition of stock.

Advantages of computer audit programs

1. Examination of data is more rapid.
2. Examination of data is more accurate
3. The only practical method of examining large amounts of data
4. Gives the auditor practical acquaintance with live files
5. Provides new opportunities to the auditor.
6. Less time consuming
7. Less paper work
8. Less disruptive to business operations
9. Overcomes in some cases a loss of audit trail
10. Relatively cheap to use once set up costs have been incurred.

Disadvantages

1. Can be expensive to setup or acquire
 2. Some technical knowledge is required
 3. Standard comp audit programs may be compatible with the variety of programming languages used in business information systems.
 4. Detailed knowledge of system & programs is required. Some auditors would dispute the need for this detailed knowledge to be gained.
 5. Difficult in obtaining computer time especially for testing.
- Use of audit s/ware raises the visibility of the auditor in the eyes of the company/organization. It makes the audit more credible.

Revision Question

1. Explain the term 'Computer Assisted Audit Techniques (CAATs)' as used in connection with computer auditing.

For further reading on Auditing Computerized Systems; System Theory and Management Information Systems by N. A. Saleemi, pages 372 - 377